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Spectrum plotting  
Graded Quiz • 15 min

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3.1 Signals as Time Dependent Data

3.2 Signal Interpolation

3.3 Audio Analysis

3.4 Convolution Filtering

3.5 Signal Frequency Analysis

Video: Signal Frequency Analysis  
1h 30m

Graded App Item: myfft  
Submitted

Quiz: Spectrum plotting  
15 min

Graded App Item: myfft  
15 min

Quiz: Filter Quality Analysis  
15 min

Graded App Item: Spectrogram  
10 min

3.6 Sampling and Aliasing

Grade received 100%

Latest Submission Grade 100%

100% higher

Spectrum plotting

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Review Learning Objectives

1. After loading the attached '.mat' file, you will have a [220500 x 1] sound signal vector 'crickets' with sampling frequency 'Fs'. When played using soundsc, it sounds quite a bit like crickets at night. But this signal is encoded with a special secret message in its magnitude spectrum.  
1 / 1 point

Submit your assignment

Try again

Implement the MagnitudeSpectrumPlot function. If you have implemented myfft and MagnitudeSpectrumPlot correctly, when viewing the MagnitudeSpectrum plot of the Fourier Transform with the third input argument col='r', letters are clearly revealed. What are they?

Receive grade

To Pass 100% or higher

Your grade 100%

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We keep your highest score

crickets

MAT File

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VU

Correct